MATH0118 Mathematics for Quantum Mechanics

Year:	2026–2027
Code:	MATH0118
Level:	7 (UG/PG)
Normal student group(s):	UG: Year 4 or Year 3 Mathematics degrees PG: MSc Mathematics
Value:	15 credits (= 7.5 ECTS credits)
Term:	1
Assessment:	80% examination, 20% coursework
Normal Pre-requisites:	MATH0056 Mathematical Methods 4 MATH0014 Algebra 3: Further Linear Algebra
Lecturer:	Prof. Rod Halburd

Course Description and Objectives

This module aims to introduce students to the mathematical framework used in quantum mechanics and teach them to apply various approximation methods, with minimal prior knowledge of quantum physics. As well as underpinning much of modern physics, quantum mechanics is important to mathematics, both as a source of important applications and as a source of ideas that have led to new directions in pure mathematics. After a brief overview of the physical motivation for quantum mechanics, we will introduce the main postulates and mathematical framework. We will solve the Schrödinger equation exactly in several important cases, such as the single-electron atom, and introduce the idea of internal symmetries and spin. Various approximation methods will be studied, including first- and second-order perturbation theory and the Wentzel, Kramers, Brillouin (WKB) approximation method.

Recommended Texts

1. S. Weinberg. Lectures on Quantum Mechanics. Cambridge University Press, 2nd edition 2015. ISBN-13 978-1107111660.
2. J.J. Sakurai and J. Napolitano. Modern Quantum Mechanics. 2nd edition. Addison Wesley 2011. ISBN 13: 978-0-8053-8291-4
3. E. Merzbacher. Quantum Mechanics. John Wiley & Sons, 3rd edition 1997. ISBN-13 978-0471887027
4. S. J. Gustafson, I. M. Sigal: Mathematical Concepts of Quantum Mechanics, Springer Universitext 3rd Edition 2020. ISBN 13: 9783030595616

Detailed Syllabus

Key experiments. Wave mechanics. Matrix mechanics. Probabilistic interpretation. Operators on vector spaces. Position and momentum operators. The Schrödinger equation. The harmonic oscillator. Creation and annihilation operators. Particle in a central potential. Spherical harmonics. General principles of quantum mechanics. Observables. Symmetries. Heisenberg uncertainty principle. Rotations, angular momentum and spin. Tensor products of representations. Addition of angular momentum. Clebsch-Gordan coefficients. Bosons, fermions and internal symmetries. The algebraic derivation of the hydrogen spectrum. Time-independent approximations. First-order perturbation theory. The Zeeman effect. The Stark effect. Second-order perturbation theory. The variational method. The Born-Oppenheimer approximation. The Wentzel, Kramers, Brillouin (WKB) approximation. Time-dependent approximations. Monochromatic perturbations. Absorption and emission of radiation. Scattering theory. Resonances.